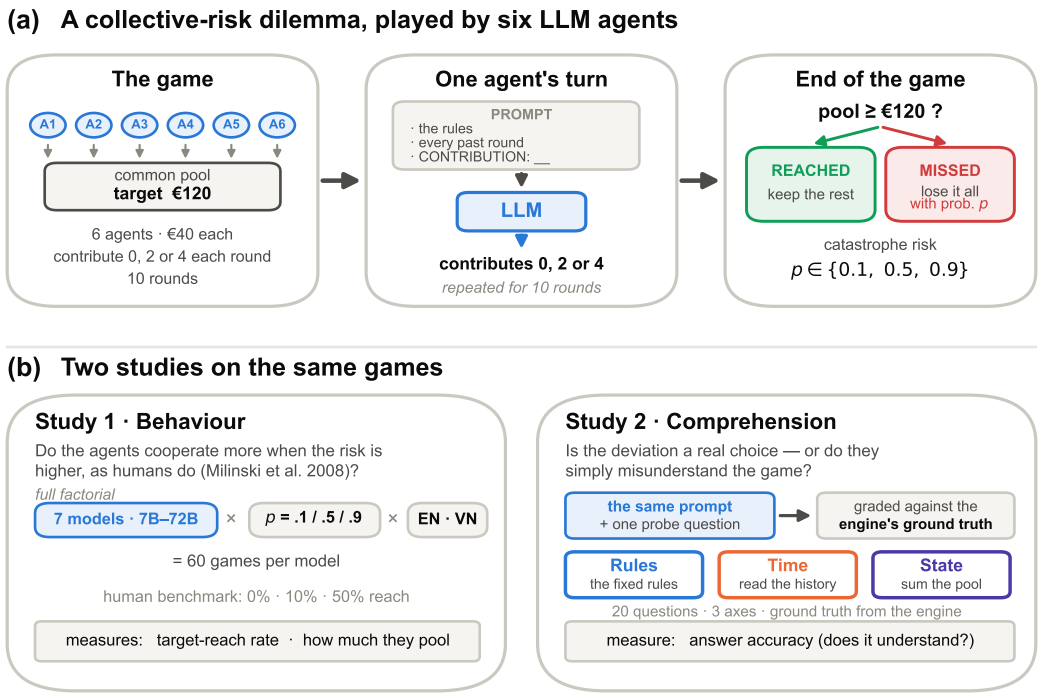
(a) A collective-risk dilemma, played by six LLM agents
The game
A1
A2
A3
A4
A5
A6
common pool
target €120
6 agents · €40 each
contribute 0, 2 or 4 each round
10 rounds
One agent's turn
PROMPT
· the rules
· every past round
· CONTRIBUTION: __
LLM
contributes 0, 2 or 4
repeated for 10 rounds
End of the game
pool ≥ €120 ?
REACHED
keep the rest
MISSED
lose it all
with prob. p
catastrophe risk
p ∈ {0.1, 0.5, 0.9}
(b) Two studies on the same games
Study 1 · Behaviour
Do the agents cooperate more when the risk is
higher, as humans do (Milinski et al. 2008)?
full factorial
7 models · 7B–72B
×
p = .1 / .5 / .9
×
EN · VN
= 60 games per model
human benchmark: 0% · 10% · 50% reach
measures: target-reach rate · how much they pool
Study 2 · Comprehension
Is the deviation a real choice — or do they
simply misunderstand the game?
the same prompt
+ one probe question
graded against the
engine's ground truth
Rules
the fixed rules
Time
read the history
State
sum the pool
20 questions · 3 axes · ground truth from the engine
measure: answer accuracy (does it understand?)